

```
!pip install openpyxl pandas matplotlib numpy
```

```
Requirement already satisfied: openpyxl in /usr/local/lib/python3.12/dist-packages (3.1.5)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl) (2.0.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.57.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.7)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_excel('Week 4.xlsx')
```

```
df.head()
```

	Id	ActivityDate	TotalSteps	Calories	AverageHeartRate	SleepHours	Unnamed: 6	Unnamed: 7
0	1503960366	3/25/2016	11004	1819	85	7.5	NaN	NaN
1	1503960366	3/26/2016	17609	2154	95	7.5	NaN	NaN
2	1503960366	3/27/2016	12736	1944	95	7.5	NaN	NaN
3	1503960366	3/28/2016	13231	1932	95	7.5	NaN	NaN
4	1503960366	3/29/2016	12041	1886	95	7.5	NaN	NaN

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 457 entries, 0 to 456
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Id                     457 non-null   int64
1   ActivityDate           457 non-null   object
2   TotalSteps             457 non-null   int64
3   Calories               457 non-null   int64
4   AverageHeartRate       457 non-null   int64
5   SleepHours             457 non-null   float64
6   Unnamed: 6             0 non-null     float64
7   Unnamed: 7             1 non-null     object
dtypes: float64(2), int64(4), object(2)
memory usage: 28.7+ KB
```

```
df = df.drop(columns=['Unnamed: 6', 'Unnamed: 7'])
```

```
df.isnull().sum()
```

	0
Id	0
ActivityDate	0
TotalSteps	0
Calories	0
AverageHeartRate	0
SleepHours	0

```
dtype: int64
```

```
df.describe()
```

	Id	TotalSteps	Calories	AverageHeartRate	SleepHours
count	4.570000e+02	457.000000	457.000000	457.000000	457.000000
mean	4.628595e+09	6546.562363	2189.452954	80.636761	7.486652
std	2.293781e+09	5398.493064	815.484523	8.048333	0.801436
min	1.503960e+09	0.000000	0.000000	72.000000	6.000000
25%	2.347168e+09	1988.000000	1776.000000	72.000000	6.800000
50%	4.057193e+09	5986.000000	2062.000000	78.000000	7.500000
75%	6.391747e+09	10198.000000	2667.000000	85.000000	8.500000
max	8.877689e+09	28497.000000	4562.000000	95.000000	8.500000

```
print("Average Steps:", df['TotalSteps'].mean())

print("Average Calories:", df['Calories'].mean())

print("Average Heart Rate:", df['AverageHeartRate'].mean())

print("Average Sleep Hours:", df['SleepHours'].mean())
```

Average Steps: 6546.562363238512
Average Calories: 2189.45295404814
Average Heart Rate: 80.63676148796499
Average Sleep Hours: 7.4866520787746165



```
df.corr(numeric_only=True)
```

	Id	TotalSteps	Calories	AverageHeartRate	SleepHours
Id	1.000000	0.138662	0.290868	0.117769	-0.287845
TotalSteps	0.138662	1.000000	0.581380	0.939800	-0.551602
Calories	0.290868	0.581380	1.000000	0.529309	-0.914281
AverageHeartRate	0.117769	0.939800	0.529309	1.000000	-0.516749
SleepHours	-0.287845	-0.551602	-0.914281	-0.516749	1.000000

```
top_users = df.groupby('Id')['TotalSteps'].mean().sort_values(ascending=False)

top_users.head(10)
```

TotalSteps	
Id	
8877689391	17417.083333
8053475328	14844.363636
6962181067	12639.714286
7007744171	12260.333333
2022484408	12174.916667
1503960366	11640.526316
2347167796	9800.066667
1644430081	9274.800000
3977333714	8663.916667
5577150313	8608.181818

dtype: float64

```
corr = df.corr(numeric_only=True)
```

```
if corr.loc['TotalSteps', 'Calories'] > 0.5:
    print("Users with higher steps tend to burn more calories.")
```

Users with higher steps tend to burn more calories.

```
corr = df.corr(numeric_only=True)
```

```
insights = []
```

```
# Steps vs Calories
if corr.loc['TotalSteps', 'Calories'] > 0.5:
    insights.append(
        "Users with higher daily steps tend to burn significantly more calories."
    )

# Steps vs Heart Rate
if corr.loc['TotalSteps', 'AverageHeartRate'] > 0.5:
    insights.append(
        "Average heart rate increases with physical activity levels."
    )

# Sleep vs Calories
if corr.loc['SleepHours', 'Calories'] < -0.5:
    insights.append(
        "Sleep duration shows a negative relationship with calories burned."
    )

# Sleep vs Steps
if corr.loc['SleepHours', 'TotalSteps'] < 0:
    insights.append(
        "Highly active users appear to sleep slightly less in this dataset."
    )

# Average Steps Insight
avg_steps = df['TotalSteps'].mean()

if avg_steps < 10000:
    insights.append(
        "Average daily steps are below the recommended 10k daily activity goal."
    )

# Print AI-generated insights
print("AI GENERATED INSIGHTS:\n")

for i, insight in enumerate(insights, 1):
    print(f"{i}. {insight}")
```

AI GENERATED INSIGHTS:

1. Users with higher daily steps tend to burn significantly more calories.
2. Average heart rate increases with physical activity levels.
3. Sleep duration shows a negative relationship with calories burned.
4. Highly active users appear to sleep slightly less in this dataset.
5. Average daily steps are below the recommended 10k daily activity goal.

Start coding or [generate](#) with AI.

